

In situ measurements of atmospheric methane at GAGE/AGAGE sites during 1985–2000 and resulting source inferences

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[1] Continuous measurements of methane since 1986 at the Global Atmospherics Gases Experiment/Advanced Global Atmospherics Gases Experiment (GAGE/AGAGE) surface sites are described. The precisions range from approximately 10 ppb at Mace Head, Ireland, during GAGE to better than 2 ppb at Cape Grim, Tasmania, during AGAGE (i.e., since 1993). The measurements exhibit good agreement with coincident measurements of air samples from the same locations analyzed by Climate Monitoring and Diagnostics Laboratory (CMDL) except for differences of approximately 5 ppb before 1989 (GAGE lower) and about 4 ppb from 1991 to 1995 (GAGE higher). These results are obtained before applying a factor of 1.0119 to the GAGE/AGAGE values to place them on the Tohoku University scale. The measurements combined with a 12-box atmospheric model and an assumed atmospheric lifetime of 9.1 years indicates net annual emissions (emissions minus soil sinks) of 545 Tg CH₄ with a variability of only ± 20 Tg from 1985 to 1997 but an increase in the emissions in 1998 of 37 ± 10 Tg. The effect of OH changes inferred by *Prinn et al.* [2001] is to increase the estimated methane emissions by approximately 20 Tg in the mid-1980s and to reduce them by 20 Tg in 1997 and by more thereafter. Using a two-dimensional (2-D), 12-box model with transport constrained by the GAGE/AGAGE chlorofluorocarbon measurements, we calculate that the proportion of the emissions coming from the Northern Hemisphere is between 73 and 81%, depending on the OH distribution used. However, this result includes an adjustment of 5% derived from a simulation of the 2-D estimation procedure using the 3-D MOZART model. This adjustment is needed because of the very different spatial emission distributions of the chlorofluorocarbons and methane which makes chlorofluorocarbons derived transport rates inaccurate for the 2-D simulation of methane. The 2-D model combined with the annual cycle in OH from *Spivakovsky et al.* [2000] provide an acceptable fit to the observed 12-month cycles in methane. The trend in the amplitude of the annual cycle of methane at Cape Grim is used to infer a trend in OH in 30°–90°S of $0 \pm 5\%$ per decade from 1985 to 2000, in qualitative agreement with *Prinn et al.* [2001] for the Southern Hemisphere.

INDEX TERMS: 0322 Atmospheric Composition and Structure: Constituent sources and sinks; 0325 Atmospheric Composition and Structure: Evolution of the atmosphere; 0315 Atmospheric Composition and Structure: Biosphere/atmosphere interactions; 1610 Global Change: Atmosphere (0315, 0325); **KEYWORDS:** methane measurements, methane modeling, tropospheric methane, methane trends

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1. Introduction

[2] Methane (CH₄) is an important trace constituent of the Earth's atmosphere, being both radiatively and chemically active. Much has been learned about its behavior in the past 2 decades. Several systematic observational programs have contributed to the documentation of the increasing global abundance of atmospheric methane over this period [e.g., *Dlugokencky et al.*, 2001; *Worthy and Ernst*, 1999; *Mahieu et al.*, 1997; *Matsueda et al.*, 1996; *Aoki et al.*, 1992; *Khalil and Rasmussen*, 1990; *Brunke et al.*, 1990; *Blake and Rowland*, 1988; *Fraser et al.*, 1986]. While there has been an overall increase of 13% in methane abundance between 1978 and 1999, the growth rate has been declining [e.g., *Dlugokencky et al.*, 1998; *Matsueda et al.*, 1996;

Khalil and Rasmussen, 1993; Steele *et al.*, 1992], but accompanied by large interannual variations [Dlugokencky *et al.*, 2001, 1996, 1994a]. Natural processes such as volcanic eruptions seem capable of causing such interannual variations in the global methane growth rate. The eruption of Mount Pinatubo in June 1991 may have acted to reduce tropical tropospheric hydroxyl radical (OH) concentrations, causing an increased methane growth rate during 1991 and early 1992 [Dlugokencky *et al.*, 1996]. Cooler temperatures resulting from the Pinatubo eruption likely acted to reduce methane emissions from natural wetlands in the Northern Hemisphere during late 1992 and 1993 [Hogan and Harriss, 1994], leading to reduced methane growth rates. Recently, Camp *et al.* [2001] have identified a small variation in tropospheric methane that is linked to the interannual variability of stratospheric ozone.

[3] The directly observed record of atmospheric methane has been extended far into the past by the analysis of air extracted from polar firn and ice cores [e.g., see Chappellaz *et al.*, 2000, and references therein]. In glacier eras, CH₄ values reach as low as 350 parts per billion (ppb), rising to about 700 ppb during the warmer interglacial periods [e.g., Stauffer *et al.*, 1988; Raynaud *et al.*, 1988; Chappellaz *et al.*, 1990], indicating clearly a link between climate and the atmospheric methane burden. During the late preindustrial Holocene (1000–1800 A.D.) the global average CH₄ was 695 ppb (varying by about 40 ppb in unison with climate variations), indicating that in the past 200 years, atmospheric CH₄ levels have increased by a remarkable 150% [Etheridge *et al.*, 1998]. The high time resolution and accurate dating of the Antarctic Law Dome ice core and firn air CH₄ record, and its good linkage to modern systematic measurements, both confirm the declining growth rate of atmospheric methane since about 1981, as well as revealing, in particular, the pattern of growth during the twentieth century [Etheridge *et al.*, 1998]. The declining growth rate has been attributed to the stabilization of global methane emissions since the early to middle 1980's [Etheridge *et al.*, 1998; Dlugokencky *et al.*, 1998], a view supported by the trend of atmospheric $\delta^{13}\text{C}$ -CH₄ measured at Cape Grim, Tasmania, over the period 1978–1995 [Francey *et al.*, 1999].

[4] While accurate knowledge about present day and past trends in the atmospheric methane burden is vital, the ultimate goal of many investigations is the elucidation of the global methane budget, and how it has evolved. One key element in this quest has been the determination of the global distribution of atmospheric methane. Its spatial gradients, seasonal variations, and the latitudinal dependence of seasonal variation are now quite well characterized [e.g., see Dlugokencky *et al.*, 1994b, and references therein]. At many sites, particularly in the extensive cooperative network operated by National Oceanic and Atmospheric Administration/Climate Monitoring and Diagnostics Laboratory (NOAA/CMDL), methane measurements are made by the collection of air samples in flasks approximately weekly since 1983, with return to a central laboratory for analysis [e.g., Dlugokencky *et al.*, 1994b; Steele *et al.*, 1987]. At a more limited number of sites, in situ measurements of methane have been made, sometimes in conjunction with a flask sampling program [e.g., Brunke *et al.*,

1990; Aoki *et al.*, 1992; Khalil and Rasmussen, 1993; Dlugokencky *et al.*, 1995; Worthy *et al.*, 2000].

[5] Some features of the global methane budget are now relatively well known. The primary removal process is destruction by OH in the atmosphere. The annual loss of methane by this process can be estimated, using known reaction kinetics and the globally weighted average tropospheric OH derived from knowledge of atmospheric 1,1,1-trichloroethane [Prinn *et al.*, 1995, 2001]. Knowledge of the atmospheric burden and the growth rate of methane allow the estimation of annual emissions. The major terms which comprise the present-day global budget of atmospheric methane have been identified, and their likely geographical locations have been mapped in some detail [e.g., Fung *et al.*, 1991, and references therein; Olivier *et al.*, 1996]. Nevertheless accurately partitioning the total emissions across the main categories has proven to be difficult, with large uncertainties remaining in the magnitude of global emissions from several source types [e.g., see Houweling *et al.*, 1999].

[6] These uncertainties remain large for two different reasons. In the so-called “bottom up” approach, direct measurements of methane fluxes (using methods such as flux chambers) must be scaled up by large factors, across regions where the flux rates are likely to be very heterogeneous, in space or time (or both), leading inevitably to uncertainty in estimates of regional or global emissions. The “top down” approach uses an atmospheric transport model, a representation of the OH sink process, and the measured distribution of methane in the troposphere to infer the combination of sources and sinks which best match the known constraints, using either forward or inverse techniques [e.g., Fung *et al.*, 1991; Hein *et al.*, 1997; Lelieveld *et al.*, 1998; Houweling *et al.*, 1999]. But uncertainties of such estimated fluxes remain stubbornly large, probably reflecting both insufficient observational constraints and limitations of the models. A discussion of how the top down approach might be improved is provided by Houweling *et al.* [2000]. In particular, they examine how quasi-continuous measurements of methane at some sites might impose tighter constraints on estimated methane emissions.

[7] In this paper we describe the in situ methane measurements made as part of the Global Atmospheric Gases Experiment/Advanced Global Atmospheric Gases Experiment (GAGE/AGAGE) program since 1985. The quality of the data and its use in drawing inferences about the methane budget are assessed. The high frequency of these measurements reveal patterns of short-term variations which can not be captured by flask sampling, thus providing additional important information that may help constrain estimates of methane emissions. The synchronization of these methane measurements with those of a range of other trace gases made by GAGE/AGAGE [e.g., Simmonds *et al.*, 1993; Derwent *et al.*, 1998] provides further opportunities to identify and quantify regional methane emissions as shown by Ryall *et al.* [2001] and Dunse *et al.* [2001]. However in this paper we limit our discussion to long-term changes and we examine what, if any, constraints on the global methane budget can be derived from using a 2-D model [Cunnold *et al.*, 1994] which has been constrained by simultaneous measurements of the chlorofluorocarbons (CFCs) and

CH_3CCl_3 at five surface locations. It will be shown that the inferred distribution of CH_4 emissions differs from estimates by others; 3-D model calculations are then used to interpret this discrepancy. Finally the observed CH_4 seasonal cycle is examined for implications regarding the trend in hydroxyl.

2. GAGE/AGAGE Measurement Methods

[8] Ground-based measurements of methane are reported here through December 2000. Measurement began at Mace Head, Ireland (53°N , 10°W), in February 1987; at Ragged Point, Barbados (13°N , 59°W), in November 1985; at Cape Matatula, American Samoa (14°S , 171°W) in February 1987; and at Cape Grim, Tasmania (41°S , 145°E), in May 1986. Measurements were also made at Cape Meares, Oregon (45°N , 124°W) from September 1985 until June 1989. This site was replaced by Trinidad Head, California (41°N , 124°W), from which methane measurements have been reported since October 1995. Although these methane data have appeared in various archives [e.g., *Prinn et al.*, 1998; *GLOBALVIEW-CH₄*, 1999], they have not previously been discussed in any detail. The measurement procedure is therefore described here.

[9] Methane measurements prior to August 1993 were made on modified GAGE microprocessor-controlled HP5880 instruments [*Prinn et al.*, 2000]. The modifications made in order to measure methane included the installation of a gas sampling valve fitted with a 3-cm^3 sample loop, a molecular sieve third column and a flame ionization detector. The carrier gas used for the methane channel was high purity N_2 . The flame gases were 40% O_2 in N_2 and ultra high purity H_2 . Twelve measurements of methane in dried ambient air were made at 2-hour intervals throughout the day; these were alternated with twelve measurements of air from an on-site standard tank containing air with known mole-fractions of methane and chlorofluorocarbons.

[10] Beginning in 1993 the HP5880s used in the GAGE program were progressively replaced by the modified HP5890s which are used in the AGAGE program. The front end for the HP5890 gas chromatographs was designed at Scripps Institution of Oceanography and consists of a sophisticated, custom-built, thermostatted gas sampling assembly containing flow control and column switching valves. For the methane measurements, a Carle AGC-211 gas chromatograph with a flame ionization detector (FID) was added in parallel with the HP5890s and used the same gas sampling assembly. “Zero-Air” from an Aadco Pure Air generator, and ultra high purity H_2 are used as flame gases. Air samples first pass through a Nafion membrane drier (Perma Pure) which is purged with a counter-current flow of dry zero air, then the series connected sample loops are flushed and loaded. The samples are injected through valves into a silica gel precolumn to remove CO_2 , water and less volatile contaminating gases and a molecular sieve column to resolve the CH_4 . The columns are maintained at 60°C and the sample loop size is 10 cm^3 . Samples are injected at 20-min intervals, with alternate ambient air and standard gas samples, giving a total of 36 calibrated air measurements per day. The CH_4 measurement technique was adapted from that used for analyzing flask samples [*Steele et al.*, 1987].

[11] The customary GAGE/AGAGE procedure for tying the measurements to a fixed standard is employed [e.g., *Prinn et al.*, 2000]. Specifically working standard tanks are filled with ambient air “under baseline conditions” (i.e., at times when the winds are from sectors free of local sources of pollution). These air samples are calibrated for CH_4 mole fractions based on measurements against a set of primary or secondary standards. They are then shipped to the GAGE/AGAGE sites for use over 3- to 6-month time periods before being replaced and returned to the calibrating laboratory for recalibration. Up until 1989 these standard tanks were filled at Cape Meares and mole fractions assigned by the (then) Oregon Graduate Center (OGC). From 1990 to 1996 they were filled at Cape Grim and mole fractions were assigned at Commonwealth Scientific Industrial and Research Organisation (CSIRO). After 1996 they were filled at Trinidad Head and mole fractions were assigned at Scripps Institution of Oceanography (SIO). Assignments made at OGC and at CSIRO prior to 1993 were in the methane scale maintained by OGC (e.g., for the Cape Grim data reported by *Fraser and Derek* [1995]). Since 1993, data have been reported in the CSIRO94 CH_4 scale [*Fraser et al.*, 1999], for which scale factors have been established against each of the OGC, NOAA/CMDL and Nippon Sanso methane scales. The CSIRO94 CH_4 scale was derived from and is almost identical to the scale maintained at NOAA/CMDL. The link was established using two high-pressure cylinders of dry natural air exchanged in 1990. Since then, a further 12 cylinders (of dry natural air or CH_4 -in-zero-air) have been exchanged with results indicating a small but measurable scale difference. The implied scale factor is 1.00021 ± 0.00010 (CSIRO/CMDL), equivalent to a difference of 0.36 ppb (CSIRO94-CMDL) at 1700 ppb (R.L. Langenfelds et al., CSIRO GASLAB measurement of CO_2 , CH_4 , CO , H_2 , and N_2O by gas chromatography, manuscript in preparation, 2001) (hereinafter referred to as Langenfelds et al., manuscript in preparation, 2001). The scale factor linking CSIRO94 and OGC scales was determined from exchange of 24 high pressure cylinders of undried natural air as 0.9804 ± 0.0004 (CSIRO94/OGC), equivalent to a difference of 34 ppb (OGC-CSIRO94) at 1700 ppb [*Fraser and Derek*, 1995]. Here we report data in the scale established by Tohoku University (TU), which is based on gravimetric preparation of mixtures by Nippon Sanso, and should therefore more accurately reflect absolute CH_4 mole fraction. Its link to the CSIRO94 CH_4 scale was established using 2 high-pressure cylinders of dry CH_4 -in-zero-air. The implied scale factor was 1.0119 (TU/CSIRO94), equivalent to a difference of 20 ppb (TU – CSIRO94) at 1700 ppb (Langenfelds et al., manuscript in preparation, 2001). Every GAGE/AGAGE CH_4 unknown air measurement is calibrated by comparing its response to the mean response of the pair of measurements of the on-site working standard made immediately before and after the unknown sample.

[12] The precision of the methane measurements is routinely estimated for the AGAGE instruments. It is obtained from the repeatability of the on-site measurements of the working standards and the daily standard deviations are tabulated every day on the AGAGE database (for example, see Figure 1b where this measure of the precision of the measurements is included in the Mace Head, Ireland, data time series). The short-term (less than 1 day) precision of

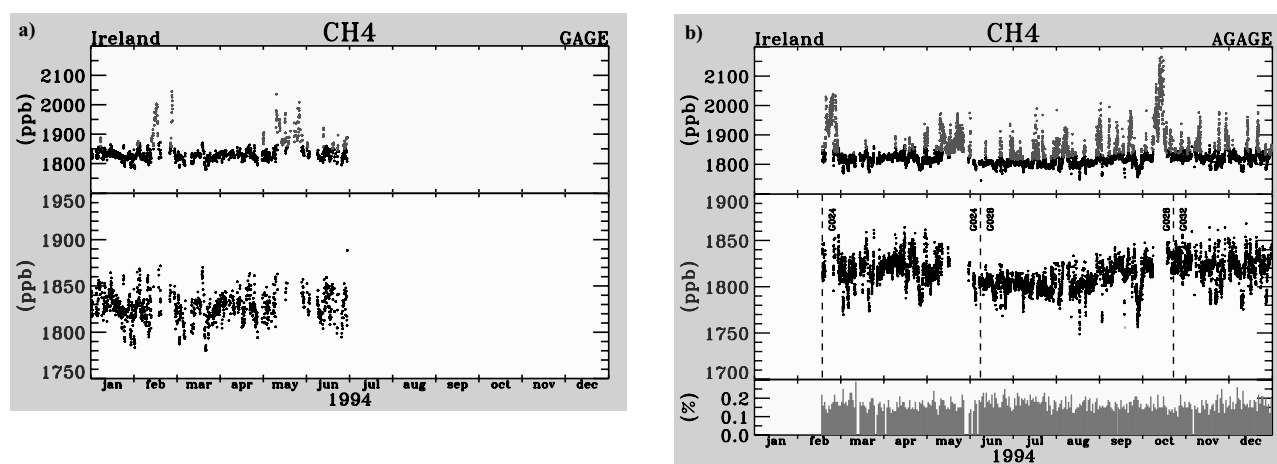


Figure 1. Methane measurements at Mace Head, Ireland, for 1994 (a) from the GAGE instrument, with a 2-hourly sampling interval and (b) from the GAGE instrument with a 40-min sampling interval. Red values indicate polluted values as designated by a statistically based pollution algorithm, standard tank changes are indicated by the dashed lines, and the blue values for AGAGE are the daily standard deviations of the measurements of the calibration air samples (i.e., measures of the instrumental precision). See color version of this figure at back of this issue.

the measurements is thus indicated to range from approximately 0.075% at Cape Grim, Tasmania, to 0.15% at Mace Head, Ireland. Because a measurement of both an air sample and a calibration gas sample is needed to produce a calibrated AGAGE measurement, this suggests that the short-term precision (one sigma) of the AGAGE methane measurements, ranges from approximately 0.1% (1.7 ppb) at Cape Grim to 0.2% (3.6 ppb) at Mace Head. The precision of the GAGE measurements can be similarly estimated and is approximately 0.3% at Cape Grim and 0.6% at Mace Head (or approximately three times worse than for the AGAGE instrument). However, in contrast to the AGAGE data, the daily precision of the measurements of the working standards is not provided as an archived output product in the GAGE database.

[13] An important issue regarding the measurements of regional pollution (which are indicated in red in Figure 1) and their use to characterize local sources of emission is the accuracy of the event maxima and whether there are any (unaccounted for) nonlinearities in the detectors. This issue is resolved for the AGAGE instruments because they contain a very accurate pressure measurement and control system allowing a range of different known sample amounts to be routinely measured. This directly provides an estimate of any nonlinearity in the system response and the estimate has been used to correct each measurement. The good agreement between the GAGE and AGAGE mole fractions during pollution events in the instrumental transition period provides confidence that any nonlinearities in the GAGE measurement system are small (no corrections for nonlinearity have been made to the GAGE measurements).

3. GAGE/AGAGE Methane Data Quality and Comparisons Against CMDL Measurements

[14] Figure 2 shows the monthly means and standard deviations of the GAGE/AGAGE time series of methane measurements. These monthly means have been computed

after removal of regional pollution data. There are gaps in the time series due for example to input value problems and the GAGE instrument at Samoa produced no data for methane or any other gas from mid-1989 until 1991 because of the destruction of the instrument by lightning. All the data are given by *Prinn et al.* [1998].

[15] There are periods of anomalously large variability at both Samoa and Mace Head, especially for the Mace Head measurements prior to 1990. The entire Barbados GAGE methane measurements (not shown) were overly variable and intermittent primarily because of switching valve problems. The Barbados GAGE methane measurements have not been included in the GAGE/AGAGE database because flask measurements were made at the same location by CMDL beginning in 1985. They provide a time series with substantially less month-to-month and year to year variability than the Barbados GAGE methane measurements and they are more consistent with the GAGE and flask measurements at the other sites. The Barbados AGAGE methane data are of high quality and are retained. The pre-1990 methane data at Mace Head are also of poorer quality, but these are retained because although there seems to be some anomalous variability, there were no CMDL flask samples there before 1990. In addition Mace Head provides data on regional pollution effects which are not found in any flask sample time series.

[16] The time of transition from the GAGE to the AGAGE instrumentation varied from site to site. At Mace Head, Ireland AGAGE measurements began in February 1994; there were approximately 4 months of overlapping data during the transition. At Ragged Point, Barbados, AGAGE measurements started in June 1996 and there were several weeks of overlapping data. AGAGE measurements at Cape Matatula, Samoa, began in August 1996; they included 1 month of overlapping data. At Cape Grim, Tasmania, AGAGE measurements started August 1993 with 17 months of overlapping measurements. Measurements at Trinidad Head, California, began in October 1995 but the

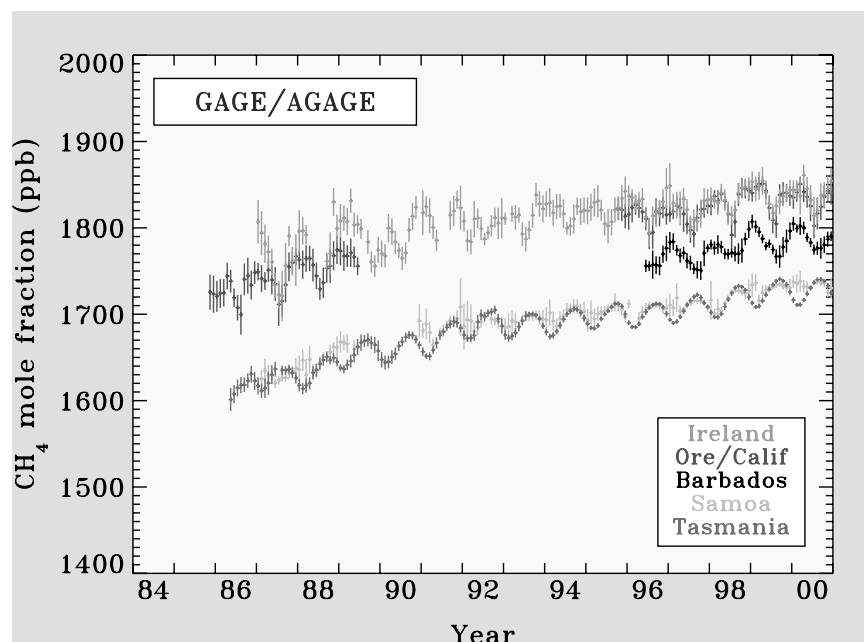


Figure 2. Monthly means and standard deviations of the baseline GAGE/AGAGE methane measurements at Mace Head, Ireland; Cape Meares, Oregon; and the follow-on site at Trinidad Head, California; Ragged Point, Barbados; Cape Matatula, Samoa; and Cape Grim, Tasmania. See color version of this figure at back of this issue.

GAGE measurements at Cape Meares, Oregon, had ceased in June 1989.

[17] A comparison of the GAGE and AGAGE data during the 17-month overlapping data period at Cape Grim (not shown but of the type shown in Figure 1) indicates several things. First the short-term (approximately 1 day) variability in the baseline values (i.e., after removal of regional pollution effects) has a standard deviation of approximately 0.1% for AGAGE and 0.3% for GAGE. There is little correlation among the short-term variations, and these standard deviations are similar to the short-term precision levels which were previously indicated. Pollution is observed more frequently in the AGAGE measurements both because of the three times higher frequency of the measurements and because of the more precisely measured baseline (see Figure 3). However, the magnitude of a typical large, narrow pollution event (the area under it) is similar. These points are illustrated in Figure 3 which shows the entire set of methane measurements at Cape Grim. Figure 3 also shows however that the width of the scatter of GAGE baseline values at Cape Grim was substantially larger during the first year of the observation which is consistent with the short term precision of the measurements being approximately 0.6% in 1986 [see also *Fraser and Derek, 1988*]. During the overlap period GAGE baseline values are approximately 0.1% larger than AGAGE values; this difference has an insignificant effect on multi-year trends. It is, at least partially, related to the poorer pollution separation that is possible for the GAGE measurements.

[18] At Mace Head, even after removal of pollution events, the GAGE and AGAGE CH_4 time series possess significant correlation even over periods of less than 1 month because of synoptic variability (Figure 1). The residual uncorrelated variability in the 4-month overlap

period is 0.4–0.6%. This is consistent with a measured short-term precision of the Mace Head AGAGE instrument of approximately 0.2% for CH_4 measurements (Figure 1) and with 0.5 to 0.6% for GAGE measurements in 1994.

[19] In 1989 the precision of the Mace Head measurements was closer to 1%. This is evident in the baseline variability, particularly on a time scale of more than 1 month (not shown). The poorer precision of the early GAGE methane measurements is a feature of the observations at all the sites in varying degrees and for varying length periods (as already noted, for example, at Cape Grim before mid-1987 and as is evident in the baseline behavior in Figure 3).

[20] A statistical procedure has been used to identify and separate out measurements which are inferred to have been influenced by regional pollution (the red dots in Figures 1 and 3). Briefly, a second-order polynomial is fitted to the subset of daily minima in any 120-day period; this provides a first estimate of the baseline. Then a standard deviation is calculated from all the deviations from this baseline in the 120-day period and values exceeding three standard deviations above the baseline are temporarily marked as polluted. The median of all the nonpolluted values in the 120-day period is determined and a Gaussian is fitted to values less than 1.5 standard deviations above the baseline. The Gaussian provides a new estimate of the standard deviation of nonpolluted data. The procedure is repeated a second time using this measure of the standard deviation. Finally, values exceeding the baseline plus three standard deviations are marked as polluted. In addition, measurements immediately preceding or following a polluted value are also marked as polluted if they exceed the baseline plus two sigma and this part of the test is continued until no more polluted values are identified. The procedure is described in more detail by

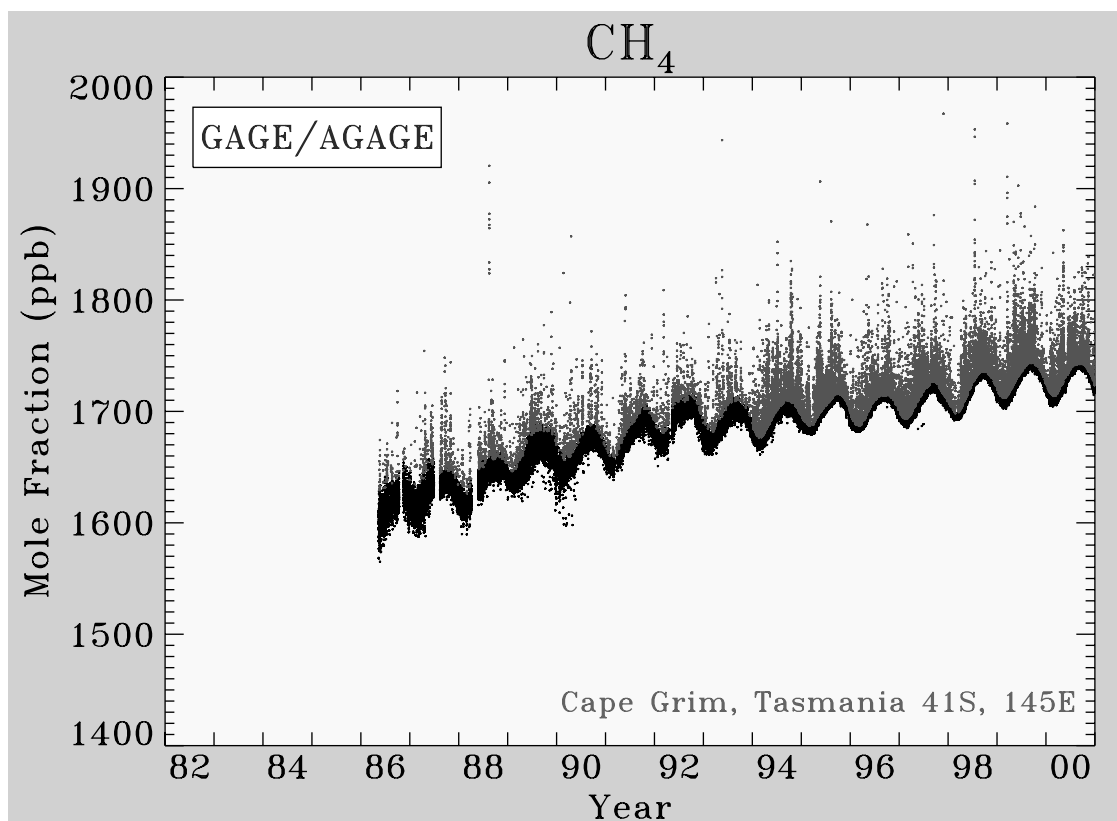


Figure 3. Every methane measurement by the GAGE and AGAGE instruments at Cape Grim, Tasmania. As in Figure 1, probably polluted values are indicated in red. The low values in early 1990 are probably related to measurement problems. See color version of this figure at back of this issue.

O'Doherty *et al.* [2001] and its excellent performance, versus trajectory type procedures, for resolving the baseline, even for the highly polluted CHCl_3 measurements, is illustrated therein. The analysis in the remainder of this manuscript is based on the baseline values only.

[21] Table 1 shows the coefficients obtained by fitting the empirical model to the monthly mean time series shown in Figure 2 where all of the coefficients a , b , d , e , and f , c_1 , and s_1 and c_2 and s_2 are determined optimally by weighting each monthly mean by the inverse of $\sigma_j^2 + \sigma_i^2$; where σ_j is the standard error of the mean for month j and σ_i^2 is the residual variance (adjusted for the month to month correlation of the residuals) based on fitting the empirical model (equation (1)) to the measurements. The P_k are Legendre polynomials of order k , and t is measured in years from the beginning of the $2N$ year interval of interest.

$$\begin{aligned} \chi(t) = & a + bNP_1\left(\frac{t}{N} - 1\right) + \frac{1}{3}dN^2P_2\left(\frac{t}{N} - 1\right) + eP_3\left(\frac{t}{N} - 1\right) \\ & + fP_4\left(\frac{t}{N} - 1\right) + c_1\cos(2\pi t) + s_1\sin(2\pi t) \\ & + c_2\cos(4\pi t) + s_2\sin(4\pi t). \end{aligned} \quad (1)$$

Here a is the mean value for CH_4 over the period being analyzed, b is the CH_4 trend, d is the rate of change of the trend, and e and f are orthogonal curvature coefficients. The c_i and s_i describe fits to the 12- and 6-month cycles in methane. Legendre polynomials are used because of their orthogonality over the $2N$ year interval which thus results

in independent estimates of the a , b , d , e , and f coefficients. For site-to-site comparability of the coefficients the fitted data employed for Table 1 have been restricted to 1 July 1986 to 31 December 2000. There were extended periods of missing data (6 months or more) at all the sites except Cape Grim and these can influence the results at individual sites. Moreover, because of the greater baseline variability of the Mace Head observations prior to 1990, coefficients are also indicated for that site after removal of the data for 1 July 1986 to 30 June 1990. The absence of several years of data at the beginning or end of any long-term record will influence the mean and the trend estimation (if these are time dependent); the coefficients for 1 July 1990 to 31 December 2000 are therefore not directly comparable to the other results and the longer period Mace Head data may provide more reliable estimates of the polynomial coefficients for the 1986 to 1999 period.

[22] Table 1b also shows comparable coefficients obtained by fitting monthly means of CMDL flask sample data (obtainable from www.cmdl.noaa.gov) at these same locations (but including Barbados in this case) with the same empirical model over the same time period. For the CMDL flask measurements the only extended period of missing data occurs because the Mace Head sampling only started in mid-1991; for this site the coefficients are given for the 1990 to 2000 period so that they may be directly compared against the GAGE/AGAGE coefficients in line 1 of Table 1a.

Table 1a. Means, Trends, Curvatures and Seasonal Cycles (Equation (1)) for GAGE/AGAGE Monthly Mean Methane Measurements for 1 July 1986 to 31 December 2000^a

Site	Extended Period of Missing Data	a_0	b	d	e	f	c_1	s_1	c_2	s_2	σ_1^2
Mace Head, Ireland	1 July 1986 through 1 July 1990	1822.9 ± 1.1	4.13 ± 0.38	−0.10 ± 0.30	0.2 ± 2.9	−0.3 ± 3.3	−11.7 ± 1.6	−3.5 ± 1.6	−3.8 ± 1.1	−5.9 ± 1.1	130
Mace Head, Ireland	1 July 1986 through 1 Feb. 1987	1810.9 ± 2.1	5.33 ± 0.53	−0.46 ± 2.9	2.8 ± 5.6	−0.1 ± 6.2	−11.6 ± 2.0	−5.7 ± 2.0	−3.9 ± 1.0	−4.7 ± 1.0	600
Cape Meares, Oregon/Trinidad Head, California	1 July 1989 through 1 Oct. 1995	1794.4 ± 2.0	7.16 ± 0.24	−0.32 ± 0.23	1.5 ± 3.1	−6.0 ± 3.8	−12.4 ± 1.0	−6.3 ± 1.0	−6.7 ± 1.0	−6.4 ± 1.0	100
Point Matatula, Samoa	1 July 1989 through 1 Dec. 1990	1691.2 ± 0.8	7.44 ± 0.21	−0.89 ± 0.12	14.4 ± 2.8	−3.2 ± 2.9	−0.7 ± 1.2	2.8 ± 1.2	0.3 ± 0.6	2.2 ± 0.6	90
Cape Grim, Tasmania	none	1685.0 ± 0.9	7.58 ± 0.21	−0.73 ± 0.12	7.9 ± 2.2	1.8 ± 2.6	5.6 ± 0.5	11.8 ± 0.5	0.4 ± 0.3	−1.9 ± 0.3	140

^a σ_1^2 is the residual variance at multi-year periods. Note that the analysis period for the first line of Mace Head data is 1 July 1990 to 31 December 2000.

[23] The empirical model results indicate differences in absolute calibrations (based on the almost continuous Samoa and Tasmania measurements) which arise almost entirely from the application of the Tohoku University scale factor of 1.0119 to the GAGE/AGAGE measurements. Apart from that there may be seen to be excellent agreement between the two sets of measurements except at Cape Meares, Oregon, for which the CMDL trend appears to be too small. Both sets of measurements show methane increasing in the Southern Hemisphere (and Barbados) at an average rate of 7–7.5 ppb/year from 1987 to 2000 but in the northernmost semihemisphere the trends are smaller. Negative values of the “ d ” coefficient indicate that the rate of increase has been decreasing about 0.5 ppb/yr each year over this 14.5-year time period. Generally positive values of the coefficient e are also evident (with some variability between sites and large error bars). This is associated with a flattening out of the upward trend in about 1991. Differences between the e and f coefficients for GAGE/AGAGE and CMDL suggest that there are some differences between the two sets of measurements over shorter time periods. Nevertheless there is excellent agreement between the seasonal cycle coefficients, c_i and s_i , for example at all the sites.

[24] More insight into the differences between the GAGE/AGAGE and CMDL data may be obtained by comparing coincident observations. Coincidences here are defined as within 2.4 hours, which is chosen to ensure that approximately 90% of the CMDL measurements have a

corresponding GAGE/AGAGE measurement during the times when both networks are operating routinely. Since the CMDL sampling procedure tries to avoid local pollution whenever possible, no problems are anticipated with using coincidence delays of up to 2.4 hours. This is confirmed by the dominance of low frequencies in 1-month power spectra of baseline Mace Head data even in months with strong pollution and in negligible differences in baseline periodicities of less than 4 months between polluted and unpolluted months.

[25] Figure 4 summarizes the coincidence results for each of the five sites in the form of monthly means and standard deviations of differences of the coincident measurements. In these calculations CMDL data were excluded if they were flagged indicating sampling or analytical problems and if they were flagged as nonbackground [Dlugokencky *et al.*, 1994b]. There are occasions when the CMDL measurements at Mace Head are evidently polluted but the AGAGE measurements are not and there are three separate AGAGE/CMDL coincidences, within a few minutes of each other, which have been found to have differences of 20–100 ppb, with the AGAGE measurements indicated to have been polluted. Thus slight differences of sampling location and measurement duration can occasionally result in large differences in the measured values. The large difference at Samoa in December 1991 is probably related to a GAGE instrumental problem. However our data retention procedure has always been to retain data unless a specific instrumental problem has been identified (see also Figure 3).

Table 1b. Means, Trends, Curvatures and Seasonal Cycles (Equation (1)) for CMDL Methane Measurements for 1 July 1986 to 31 December 2000^a

Site	Extended Period of Missing Data	a_0	b	d	e	f	c_1	s_1	c_2	s_2	σ_1^2
Mace Head, Ireland	1 July 1986 through 1 June 91	1798.6 ± 1.5	4.77 ± 0.63	−0.10 ± 0.51	−1.8 ± 5.2	−0.2 ± 5.0	−9.3 ± 1.6	−5.1 ± 1.6	−4.2 ± 1.6	−4.9 ± 1.6	80
Cape Meares, Oregon	1 April 1998 through 31 Dec. 2000	1769.6 ± 3.1	3.00 ± 1.10	−2.17 ± 0.57	−17.8 ± 8.6	−8.7 ± 6.1	−12.2 ± 1.2	−7.3 ± 1.2	−2.9 ± 1.6	−7.2 ± 1.6	50
Ragged Point, Barbados	1 July 1986 through 1 Nov. 1987	1730.4 ± 1.5	7.00 ± 0.47	−0.77 ± 0.26	11.0 ± 4.8	−2.6 ± 4.8	−10.2 ± 1.1	−8.5 ± 1.1	1.8 ± 0.9	0.2 ± 0.9	100
Point Matatula, Samoa	none	1669.8 ± 0.9	7.21 ± 0.20	−0.53 ± 0.11	6.1 ± 2.2	2.7 ± 2.5	0.9 ± 0.7	3.9 ± 0.7	0.5 ± 0.5	2.5 ± 0.5	100
Cape Grim, Tasmania	none	1664.3 ± 0.5	7.20 ± 0.13	−0.49 ± 0.07	5.8 ± 1.4	1.6 ± 1.6	5.7 ± 0.4	12.8 ± 0.4	0.0 ± 0.2	−1.9 ± 0.2	40

^aAnalysis period for the Mace Head data is 1 July 1990 to 31 December 2000.

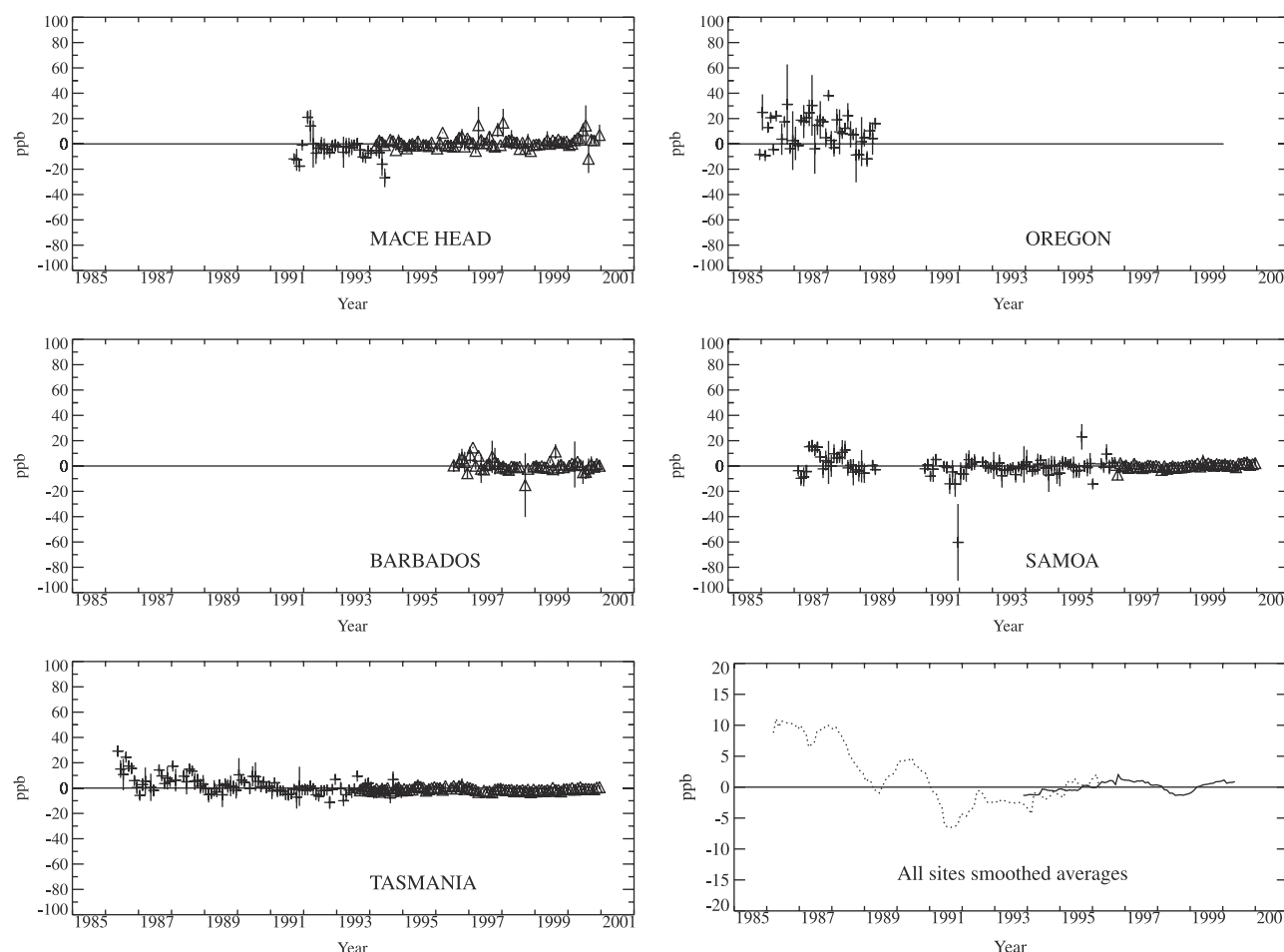


Figure 4. Monthly mean methane differences (and their standard deviations) between coincident (less than 2.4 hours apart) CMDL and GAGE/AGAGE measurements (CMDL minus GAGE/AGAGE). Pluses indicated GAGE measurement and triangles are for AGAGE. The plot in the lower right summarizes the five individual site plots. It shows the 12-month running means of the differences averaged over the five sites (the GAGE and AGAGE differences are shown as dashed and full lines, respectively). Note that the absolute calibration factor of 1.0119 has not been applied to the GAGE/AGAGE measurements here in order to make any differences more obvious.

[26] To make the differences between the two data sets clearer (Figure 4), the Tohoku University scale factor has not been applied to the GAGE/AGAGE data in this case. GAGE measurements are approximately 5 ppb lower than CMDL measurements in 1987 (9 ppb at Oregon) and they are approximately 4 ppb higher in 1991 (see Figure 4). The AGAGE measurements are, on average, approximately 1 ppb higher than the CMDL values. A similar difference of 1 ppb also applies to the average of the comparisons for Samoa and Tasmania over the entire 1987 to 1999 period, and, as can be seen in Table 1, the linear trends over this same period only differ by 0.2–0.3 ppb/year. The standard deviations of the differences in monthly means are 6 ppb for the GAGE measurements before 1991 (but 11 ppb at Oregon) and approximately 4 ppb thereafter. For AGAGE the standard deviations range from 5 ppb at Mace Head to 2 ppb at Tasmania.

[27] In summary, the GAGE/AGAGE measurements are indicated to have an overall precision ranging from 1% (17 ppb) at the start of the GAGE measurements at Mace

Head in 1987 to approximately 0.2% (3 ppb) for AGAGE measurements at Cape Grim from 1994 on. Comparisons against CMDL measurements suggest that the imprecision is similar at periods of a year or more. The AGAGE and the CMDL measurements have been shown to be in excellent agreement (apart from an absolute calibration difference of approximately 1.2%) and to have similar precisions. The poorer precision of the GAGE measurements suggests that where coincident CMDL measurements are available the principal use of the GAGE measurements is for examining the short-term variability associated with regional pollution events.

4. CH₄ Emission Estimates

[28] Attempts to quantify the various (about 10) identified sources of global methane emissions has become an increasingly active area of research over the past decade [see e.g., Prather *et al.*, 1995; Hein *et al.*, 1997; Lelieveld *et al.*, 1998]. A recent comprehensive survey of knowledge

Table 2. Methane Source Proportions In Each Semihemisphere From *Fung et al.* [1991], and Estimated Here From Measurements at the GAGE/AGAGE Sites^a

	90°N–30°N	30°N–0°	0°–30°S	30°S–90°S
<i>Fung et al.</i> [1991]	0.46	0.33	0.18	0.03
CMDL measurements using <i>Prinn et al.</i> [2000] OH	0.58 ± 0.04	0.10 ± 0.04	0.24 ± .02	0.08 ± .01
CMDL measurements using <i>Spivakovsky et al.</i> [2000] OH	0.58	0.13	0.21	0.08
CMDL measurements using MOZART OH [<i>Hauglustaine et al.</i> , 1998]	0.61	0.15	0.18	0.06

^aEstimates are made using latitudinal distributions of hydroxyl, all of which have been normalized to produce to a tropospheric lifetime for CH₄ due to OH of 8.1 years in the 2-D model. The error bars are determined from the uncertainties in the mean mole fractions of methane and in the mean mole fractions of CFC-11 and CFC-12 which are used to constrain the transport coefficients in the 2-D model. Note that the error bars do not reflect the deficiency in the modeling approach which is discussed in the text.

about methane sources is given by *Khalil* [2000]. Of the sources the best quantification probably now exists for rice paddies (approximately 50 Tg/yr [*Neue*, 1997]) and, of the major sources, perhaps the least well quantified is that from wetlands. Several of the top-down calculations of the sources have been made using 3-D models and the extensive network of CMDL methane measurement sites [e.g., *Fung et al.*, 1991]. Here however we seek to find out what can be learned from just five continuous measurement GAGE/AGAGE sites using a 2-D model which is constrained by simultaneous measurements of several other species at those locations. It is easier to incorporate these constraints in a 2-D model than in a 3-D model.

[29] The time series of methane surface measurements at the five sites can be used to estimate the global emissions of methane by applying a 2-D model to estimate the changing contents of the upper troposphere and stratosphere [see e.g., *Cunnold et al.*, 1994]. In these calculations we have used a troposphere to stratosphere exchange time of 2 years, a constant annual mean loss time for tropospheric methane due to the reaction with hydroxyl of 8.12 years, and a global lifetime (including the stratosphere) for methane of 9.11 years. The latter are based on calculations using the 2-D model together with the OH concentrations reported by *Prinn et al.* [2001]. Horizontal transport rates in the 2-D box model have been derived based on the best fit to the chlorofluorocarbon measurements [*Cunnold et al.*, 1997]. The model consists of 4 boxes each, one in each semihemisphere, in the lower troposphere, the upper troposphere, and the stratosphere (in 12 boxes in all). Seasonality in transport rates is included in this version of the box model but the transports are not allowed to vary interannually. The seasonality is discussed later when the seasonality in hydroxyl related losses is addressed.

[30] Our calculations were initialized by dividing the emission distributions from *Fung et al.* [1991] into the four semihemispheres used in the box model (see Table 2). We found that this emission distribution combined with the fluorocarbon based transport rates resulted in a substantial overestimation of the observed latitudinal gradient of methane at the GAGE/AGAGE sites. In fact in order to obtain the observed differences between the measurements at the five AGAGE sites, much more of the global methane source would have to originate in the Southern Hemisphere than in the *Fung et al.* [1991] estimates (see Table 2). The issue of the source distribution will be discussed shortly but we first estimate the global annual emissions of methane (which in our calculations are insensitive to the latitudinal distribution of the emissions).

[31] Figure 5 shows the annual emission estimates derived using the GAGE/AGAGE data, the 2-D model, and the fixed emission distribution which produces the best fit to the latitudinal distribution. Also shown in Figure 5 are the annual emission estimates derived using the CMDL data in place of the GAGE/AGAGE data. The results indicate that, as previously pointed out by *Dlugokencky et al.* [1998], methane emissions have been approximately constant from 1984 to 1996. The net emissions (emissions minus the soil sink) are estimated to be approximately 545 Tg CH₄/year and the atmospheric burden of CH₄ in 2000 is estimated to be approximately 4900 Tg. This is 6% larger than the 3-D model based estimate for 1992 by *Lelieveld et al.* [1998], of which approximately 3.5% is associated with the increases of CH₄ from 1992 and 2.5% is inferred to be due to model differences. There are indications that the emissions may have varied from year to year by ± 20 Tg and that they were 37 ± 10 Tg higher in 1998. The latter result has been discussed by *Dlugokencky et al.* [2001] and ascribed primarily to changes in wetland emissions.

[32] The uncertainties in these emission estimates result from a combination of random and, more importantly, systematic sources of error. The error bars in Figure 5 are derived from the root-mean-square differences between the best-fit box model results for each semihemisphere and the measurements at the individual sites. Figure 4 showed globally averaged differences between the GAGE and CMDL measurements; these differences are reflected in the differences in the emission estimates for 1988 and 1990, for example. The larger error bars prior to 1991 are indicative of the poorer precision of the GAGE measurements at that time.

[33] Sources of systematic uncertainty include the absolute calibration of the measurements, the inference of the global content of methane using the box model and surface measurements at just four locations, the atmospheric lifetime of methane, and the possibility of a long-term trend in hydroxyl (these calculations assumed there was none). The 95% confidence limits on the global atmospheric lifetime of methane have been estimated by *Prinn et al.* [2001] as ±2.4 years approximately based on the uncertainty in deriving OH concentrations from methyl chloroform measurements. This would lead to an almost uniform uncertainty in (net) CH₄ emissions of approximately ±70 Tg; it is expected to be the largest source of uncertainty in the CH₄ emissions.

[34] There have been model calculations and estimates based on CH₃CCl₃ measurements, of decadal changes in OH [*Krol et al.*, 1998; see also *Prinn et al.*, 1995]. The most recent estimate from CH₃CCl₃ measurements imply an

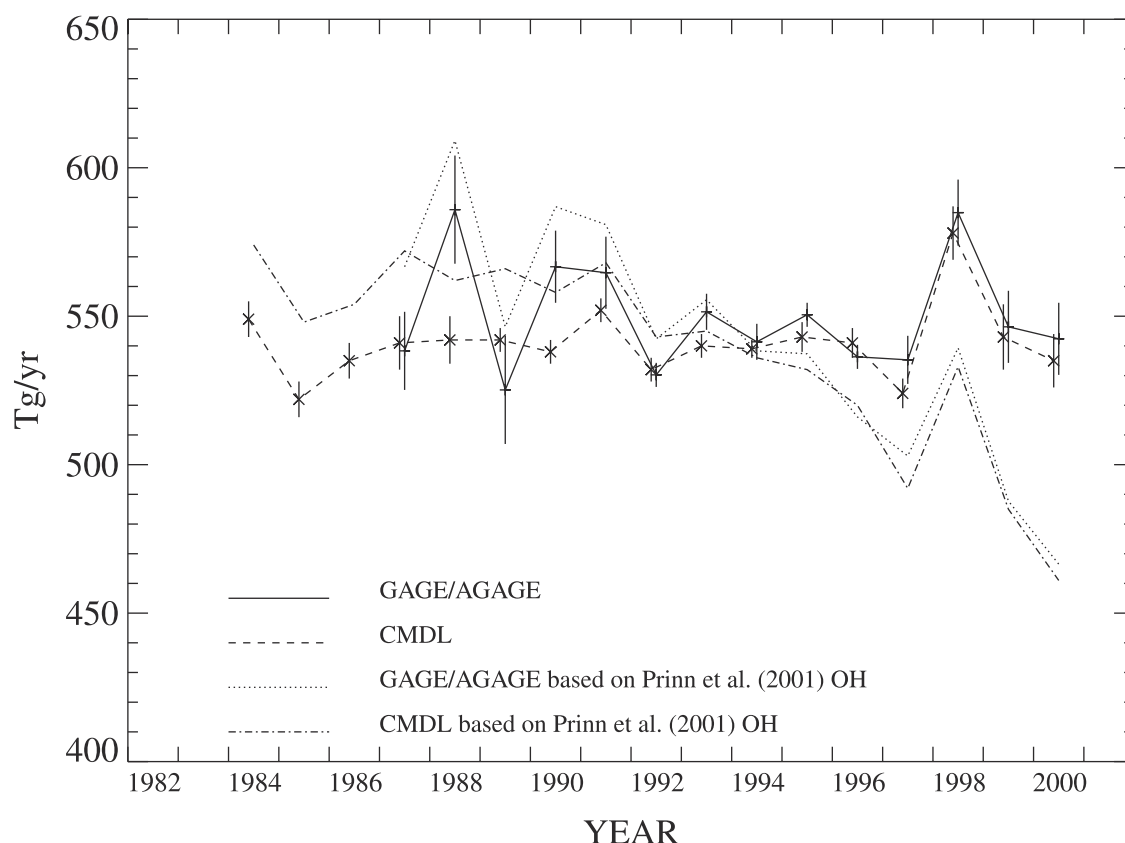


Figure 5. Calculated global annual emissions of methane and one sigma error bars based on the GAGE/AGAGE and CMDL measurements. Slight time offsets are used to keep the error bars separated. In these calculations, a tropospheric lifetime of methane due to destruction by hydroxyl equal to approximately 8.1 years has been used. The dotted and dash-dotted lines indicate the emission estimates derived from the data if the OH changes discussed by *Prinn et al.* [2001] are used.

increase of OH from 1978 to 1992 and a decrease since then [*Prinn et al.*, 2001]. Figure 5 shows that this OH scenario would result in approximately 20 Tg/yr additional methane emissions from 1983 to 1990 and approximately 20 Tg/yr emissions fewer emissions from 1995 to 1998 compared to the calculation with constant OH.

[35] The latitudinal distribution in the methane emissions has been calculated as that which provides a best fit of the model, using the inferred global emissions shown in Figure 5, to the time series of the monthly means of the GAGE/AGAGE and CMDL measurements. The procedure was previously applied to N_2O measurements as described by *Prinn et al.* [1990]. For the methane analysis, 12-month means of the data and of the modeled values were used in this study; the simulation of the seasonal cycle in methane is a separate issue which will be discussed later. Because of the large quantity of missing GAGE data in the Northern Hemisphere (and the questionable baseline in the pre-1992 measurements at Mace Head), the GAGE data do not provide as robust and consistent an estimate of the source distribution as do the CMDL measurements. However in the mean the AGAGE measurements indicate a source distribution similar to that from the CMDL data.

[36] The inferred mean source distribution (shown in Table 2) has small error bars but it is clearly very different from that of *Fung et al.* [1991]. For example, the *Fung et al.*

[1991] emission distribution when used in the 2-D model gives 90°N – 30°N /30°S–90°S semihemisphere differences of 168 ppb versus the observed value of 120 ppb. The quantifiable uncertainties in our estimates are based on the uncertainties in the multiyear estimates of the means at each site (approximately 1 ppb; see Table 1) combined with similarly derived uncertainties in the transport rates deduced from the CCl_3F and CCl_2F_2 measurements. To test the effect of possible uncertainties in the latitudinal distribution of OH on the inferred distribution of CH_4 emissions, we also ran the box model using latitudinal distributions of OH from *Spivakovsky et al.* [2000] and from *Hauglustaine et al.* [1998]. The latter distributions were however first multiplied by factors to yield a tropospheric lifetime of CH_4 due to OH of 8.1 years when used in our 2-D model. Adjustment of these OH distributions was needed both because of the different lifetimes for CH_3CCl_3 in their models compared to *Prinn et al.* [2001] and because of the different effective temperatures resulting from their OH distributions within the tropospheric boxes of the 2-D model. Table 2 indicates that the inferred distribution of CH_4 emissions has some sensitivity to the assumed OH distribution.

[37] *Hein et al.* [1997], using an inverse 3-D model and an OH distribution similar to that of *Hauglustaine et al.* [1998], inferred semihemispheric source allocations that were within 4% of those obtained by *Fung et al.* [1991].

Table 3. MOZART-1 for CH₄ and CCl₃F and CH₃CCl₃ Annual Means in the Semihemispheres and at the Grid Boxes Corresponding to the AGAGE Measurement Sites

	90–30°N	30°N–0	0–30°S	30–90°S
<i>CH₄, ppb</i>				
Semihemispheric averages 1000–500 mbar	1842	1789	1750	1742
Surface semihemispheric averages	1880	1807	1750	1743
Ireland/California, Barbados, Samoa, Tasmania after “filtering” for regional pollution	1859/1852	1788	1743	1743
<i>CCl₃F, ppt (Based on 6 Months)</i>				
Semi-hemispheric averages 1000–500 mbar	207.4	198.5	190.2	186.1
Surface semi-hemispheric averages	213.6	200.3	188.8	185.7
Ireland/California, Barbados, Samoa, Tasmania after “filtering” for regional pollution	209.4/206.3	201.2	189.0	185.8
<i>CH₃CCl₃, ppt</i>				
Semihemispheric averages 1000–500 mbar	170.0	149.6	133.0	127.4
Surface semihemispheric averages	181.8	153.7	130.5	126.7
Ireland/California, Barbados, Samoa, Tasmania after “filtering” for regional pollution	176.4/173.6	151.2	131.2	127.0

The next section discusses reasons, in addition to OH differences, that a 2-D model, with transport constrained by CFC measurements, can indicate a CH₄ source distribution quite different from that of *Fung et al.* [1991] or *Hein et al.* [1997]. In fact, the several 3-D model calculations which have been made for methane can be used to assess the limitations of any limited resolution 2-D model for estimating the emissions of methane and hence of other gases. Our 2-D model in particular is being used to estimate the sources of the many species which are being measured at the AGAGE sites [Prinn *et al.*, 2000]. An understanding of the limitations of our modeling/estimation procedure is therefore important (e.g., for CHCl₃ [O’Doherty *et al.*, 2001]).

5. MOZART 3-D Comparisons

[38] A possible explanation for the difference between 3-D models and our 2-D model semihemispheric methane source estimates is that, because of the very different spatial distribution of methane and chlorofluorocarbon sources, it is inappropriate to apply the same 2-D transport rates to methane and chlorofluorocarbons. Alternatively stated, the methane observations at the GAGE/AGAGE surface measurement locations may bear a different relationship to the semihemispheric means for 1000–500 mbar (a 2-D model grid box) than do the chlorofluorocarbons and methyl chloroform.

[39] To test this hypothesis, calculations have been made with the NCAR global chemical transport model (CTM) MOZART-1 model [Brasseur *et al.*, 1998; Hauglustaine *et al.*, 1998], in which the meteorological fields from the NCAR Community Climate Model (CCM2) were used to advect chemical species. Specifically, we discuss the results for the MOZART grid points closest to GAGE/AGAGE measurement sites for methane from the last year of the CH₄ calculation reported by Hauglustaine *et al.* [1998] and the results from the second year of calculation for chlorofluorocarbon 11 (CCl₃F) and for methyl chloroform (CH₃CCl₃). The emphasis is on the relationship between surface grid point values, especially at the GAGE/AGAGE sites, and semihemispheric averages for 1000–500 mbar. The CCl₃F source used was 3.31×10^5 kg/yr with a distribution similar to that in the GEIA database for 1985 (Global Emission

Inventory Activity, available at <http://www.geiacenter.org/Projects/chlorofluorocarbons/>) [McCulloch *et al.*, 1994]. The MOZART methane calculations were made with a net source of 455 Tg/yr (469 Tg emissions minus 14 Tg soil sink) and a tropospheric lifetime of 9.9 years as described by Hauglustaine *et al.* [1998].

[40] The MOZART model results for CH₄ and CCl₃F are summarized in Table 3 (results from just the last 6 months of the calculations for CCl₃F have been used to eliminate any residual effects of initialization). GAGE/AGAGE means are typically reported after removal of regional pollution effects. Unfortunately MOZART output was only available once per day so it did not seem useful to apply the AGAGE algorithm to the MOZART results. As an alternative approach we examined the MOZART monthly median values in the AGAGE site grid boxes and monthly means in neighboring grid boxes in the direction opposite to that in which pollution effects are most likely. If the medians and the neighboring grid box values were all similar, these values were assumed to represent the background at the sites. This test indicated for example that Melbourne, Australia, emissions were not detectable in the Cape Grim grid box, because of the relatively coarse spatial resolution of the model. For Mace Head pollution effects were fairly small and either the monthly medians in the Mace Head grid box or the means in neighboring Atlantic grid boxes could be used with only minor errors. On the other hand, MOZART clearly shows the effects of regional inputs along the West Coast of the United States for CCl₃F and CH₃CCl₃, although not for CH₄. Grid boxes immediately west of Trinidad Head (and Cape Meares) are 1 and 3% respectively lower for CCl₃F. These westward boxes which are all located off the West Coast of the United States all have similar concentrations. We use the MOZART values there, which correspond approximately to the thirty-eighth percentile of the distribution of values in the Trinidad Head grid box.

[41] The application of the 2-D model to GAGE/AGAGE measurements is based on the assumption that the measurements are typical of the corresponding semihemisphere in 500- to 1000-mbar, or at least that percentage differences are not species dependent. Table 3 examines this assumption using the MOZART calculations. The results shown in Table 3 indicate that the 90°N–30°N 1000- to 500-mbar

values are closer to the grid box values “with pollution removed” than they are to the semihemispheric surface averages. Averages in 1000- to 500-mbar semihemispheres in the Southern Hemisphere are a little higher than the surface averages because transport into the Southern Hemisphere primarily occurs in the upper troposphere. Values of CCl_3F (and CH_3CCl_3 but not of CH_4) for the Barbados grid box are however higher than 1000- to 500-mbar averages by more than 1%. In the MOZART model this is mostly due to the influence of European releases. *Prather et al.* [1987] also show high values in the vicinity of Barbados but they resulted from United States releases. The implication of these models is that surface measurements at Barbados may be significantly biased relative to the semihemispheric mean. This might not affect the inferences of the source distribution of the gases except that methane does not seem to be similarly biased. The resulting bias on the derivation of the transport rates for the 2-D model is potentially removable if CFC measurements at Guam (13°N , 145°E), where CMDL measurements of CH_4 but not of the CFCs are available, could be averaged with the Barbados measurements. MOZART results indicate that the effect of the Barbados bias would then be mostly removed. Table 3 also shows MOZART CH_3CCl_3 calculations; it is evident that the behavior of CH_3CCl_3 is similar to that of CCl_3F . This is not surprising given the fairly similar emission distributions of these compounds.

[42] We can now test how accurately the 2-D model simulates the (MOZART) CH_4 emission distribution using the MOZART OH and the MOZART model calculations of CCl_3F and CH_4 . In this test the 2-D model is initialized with a latitudinal distribution of CCl_3F similar to the initial distribution in MOZART and the transport rates are inferred based on matching the last 6 months of the MOZART semihemispheric CCl_3F values for 1000- to 500-mbar (as in Table 3). This yields an interhemispheric transport time of 9.0 months. Table 4 shows CH_4 annual means in these four regions calculated by the 2-D model. These are then compared against the corresponding CH_4 results from MOZART. Also shown are the adjustments to MOZART's CH_4 emission distribution which would be needed for the 2-D model to produce the MOZART CH_4 distribution shown in row 2 of Table 4. Thus, for example, MOZART's 44% of the emissions in 90°N – 30°N would have to be increased to 60%, and there would have to be a decrease from 36 to 15% in 30°N – 0°N . Thus for the 2-D model to simulate the MOZART CH_4 results, a substantial proportion of the emissions would have to be moved from 30°N – 0°N to 90°N – 30°N and 5% of the emissions would have to be moved from the Northern Hemisphere into the Southern Hemisphere. This strongly suggests that it is not appropriate to use the same transport coefficients in the 2-D model for CCl_3F and CH_4 , presumably because of the very different emission distributions of these gases. Note that the implied errors in inferring the MOZART CH_4 emission distribution using the 2-D model are similar to the differences between the *Fung et al.* [1991] emission distribution and that inferred using the 2-D model applied to the CMDL and GAGE/AGAGE CH_4 and CCl_3F measurements.

[43] If the MOZART values of CCl_3F and CH_4 at the GAGE/AGAGE sites were to be used in place of the semihemispheric averages in 1000–500 mbar, i.e., a more real-

istic situation, there is no change (within 0.01) of the inferred relative proportion of the emissions in the Northern Hemisphere versus the Southern Hemisphere no matter whether Barbados or Barbados/Guam averages are used. However there are significant redistributions of the emissions within the hemispheres with more of the emissions moving to the 30° – 90° semihemispheres. Thus we can conclude that the 2-D model procedure is unreliable for estimating how the CH_4 emissions are distributed within each hemisphere, but it is fairly reliable for estimating the proportion of emissions averaged over the Northern hemisphere relative to those averaged over the Southern Hemisphere.

[44] Table 2 indicates that the estimates of the Northern Hemisphere proportion of methane emissions varies from 68 to 76% depending on which of several latitudinal distributions of OH is used (see Table 5). The MOZART/2-D modeling comparison suggests that the 2-D model is not accurate enough to discriminate between the several OH distributions of Table 2. Nor is there strong independent evidence for choosing among these OH distributions. Therefore it is concluded that although our 2-D modeling procedure seems to have a bias of 5% in assigning the proportion of the CH_4 emissions in each hemisphere, the effect of the uncertainty in the OH distribution on inferring the proportion of the emissions in the two hemispheres is equally as large.

[45] It is not really surprising that results from a 2-D model with limited spatial resolution are apparently sensitive to the subgrid-scale distribution of the emissions of the long-lived gases, for example, the longitudinally strongly varying methane emissions in 0° – 30°N which maximize over Asia. The calculations described have documented the magnitude of this sensitivity. Thus the 2-D model using transport rates constrained by observations of anthropogenically produced gases, such as the CFCs, only produces accurate results on the spatial distribution of emissions for gases with fairly similar emission distributions. This condition is generally satisfied for gases whose sources are mostly anthropogenic. For the subset of other gases useful information on the hemispheric emission ratios can be obtained with the 2-D modeling/estimation procedure if the errors described are likely to be small relative to other sources of uncertainty. This is relevant to the application of the model to other species measured at the AGAGE sites (e.g., CHCl_3 [*O'Doherty et al.*, 2001]). Three-dimensional models may be favored for making emissions estimates but they too have their limitations including the accuracy with which they can represent transport and their ability to simulate point measurements at the bottom of the boundary layer [e.g., *Houweling et al.*, 2000]. Therefore the modeling of simultaneous measurements of a number of long-lived gases is a useful approach. A 2-D model has the advantage that it is more easily tuned to obtain the transport parameters to provide a best fit to all the measurements. Thus a combination of 2-D and 3-D modeling similar to that which we have described would seem to represent the best analysis approach.

6. Methane Seasonal Cycle

[46] The seasonal cycle in methane is well resolved in the GAGE/AGAGE measurements at all the sites; this is

Table 4. 2-D Model Calculation With Transport Parameters Chosen to Simulate the MOZART Semihemispheric Means of CCl₃F for Months 7–12 for 1000–500 mbar and Using the MOZART CH₄ Emission and OH Distributions^a

	90°N–30°N	30°N–0°	0°–30°S	30°S–90°S
CH ₄ semihemispheric mole fractions (ppb) for 1000–500 mbar from 2-D calculation	1834	1806	1750	1732
As above, but from the MOZART results	1842	1789	1750	1742
Adjustment in proportionate latitudinal distribution of emissions required to produce the MOZART result using the 2-D model	0.16	–0.21	0.02	0.03

^aRow 1 shows the 2-D calculated semihemispheric annual means for CH₄, and row 2 shows the corresponding results from the 3-D MOZART model. Row 3 shows the change in the latitudinal distribution of CH₄ emissions which is required for the 2-D calculation to yield the row 2 semihemispheric values.

evident from the statistical significance of the c_i and s_i values shown in Table 1. Moreover, the values are in good agreement with those from the CMDL measurements. The seasonal cycle in methane offers a good test of the ability of models to simulate the combined effects of atmospheric transport, OH variations, and the seasonality of methane emissions [Fung *et al.*, 1991]. Following the recent inferences of long-term changes in OH by Prinn *et al.* [2000], it is useful to examine long-term trends in the methane seasonal cycle in the Southern Hemisphere and to update the results from Dlugokencky *et al.* [1997]. The seasonal cycles, with seasonal minima in summer, are illustrated in Figure 6. The transport coefficients in the box model are seasonally varying [see Cunnold *et al.*, 1983]; however in order to simulate the observed 12-month cycles in CCl₃F and CCl₂F₂, the annual cycles in the transport parameters have been adjusted, primarily in phase, versus those given by Cunnold *et al.* [1983]. The annual cycles in transport result in simulated annual cycles in CCl₃F and CCl₂F₂ which are in excellent agreement with those observed, except at Barbados. In a model having just four boxes within the lower troposphere, it is not possible to make reasonable adjustments to the transport rates to simulate the annual cycles in all four semihemispheres simultaneously. The calculated annual cycles at Barbados due to transport are of the correct amplitude but they are approximately 180° out of phase with those observed (at least for CCl₃F and CCl₂F₂).

[47] The simulation of the observed 12-month components of the methane seasonal cycles has been examined for the relative roles of transport and chemistry and the relationship between seasonal cycles at the surface and those in the lower troposphere. Calculations have been made for no OH seasonal cycle and for the modified Spivakovsky *et al.*

[2000] OH distribution and seasonal cycles (Table 5). One set of calculations also included the use of MOZART's seasonal cycles in methane emissions.

[48] The seasonal cycle results are illustrated in Figure 6. The results for no OH seasonal cycle (dash-dotted lines) indicate that the seasonal cycles in transport contribute significantly to the observed cycles at all sites except Tasmania as shown in previous studies [Fraser *et al.*, 1986; Fung *et al.*, 1991]. Note the large transport induced seasonal cycle at Barbados. It is exactly in phase with the observed methane cycle and it thus results in a substantial overprediction of the observed cycle there because, as indicated above, the transport cycle, based on the CFC measurements, possesses the incorrect phase. Figure 6 shows that overall the model simulates the 12-month cycles in methane reasonably well at all the other sites, particularly if the seasonal cycles in emissions (from MOZART) are excluded. Note that the observational results include more than just the 12-month cycles. The Samoa result, in particular, is noteworthy because it indicates the cycle there is produced by strongly offsetting roles of transport and chemistry; there is also an indication that the MOZART seasonal cycle in emissions in 0°–30°S may not be correct.

[49] Because of the short timescale, the applicability of the seasonal cycles calculated with the limited resolution 2-D model to surface measurements might be considered more questionable than the model's representativeness for the annual mean concentrations at each site. This issue is addressed in part by the MOZART results shown in Figure 7. It indicates that the amplitude of the surface-averaged seasonal cycle exceeds that in the 1000- to 500-mbar semihemispheres (in the Northern Hemisphere), because of the annual cycle in emissions. The MOZART annual cycles at

Table 5. Annual Means and 12-Month Cycle Amplitudes of Hydroxyl (10^5 cm^{-3}) Based on Prinn *et al.* [2000], Spivakovsky *et al.* [2000], and Hauglustaine *et al.* [1998] by Semihemisphere^a

		90°N–30°N	30°N–0°	0°–30°S	30°S–90°S
Prinn <i>et al.</i> [2001]	annual means	5.8, 5.2	12.9, 11.4	15.3, 3.2	5.5, 5.7
Normalized Spivakovsky <i>et al.</i> [2000]	annual means	6.5, 4.9	14.3, 8.6	14.7, 9.8	5.7, 5.3
		6.2, 4.7	2.9, 1.7	2.9, 2.0	4.6, 4.2
	12 mo. cycle amplitude				
Hauglustaine <i>et al.</i> [1998]	annual means	8.0, 7.0	15.0, 11.0	13.0, 0.0	4.0, 4.0
		7.0, 6.5	4.0, 3.5	2.0, 2.5	3.0, 2.0
	12-month cycle amplitude				

^aThe Spivakovsky *et al.* [2000] concentrations have been multiplied by the factor 0.816 so as to produce tropospheric lifetimes due to hydroxyl in our model for methane and CH₃CCl₃ equal to those obtained using the Prinn *et al.* [2001] values. Values in 1000- to 500-mbar region are followed by values in the 500- to 200-mbar region.

Observations & 2D Results

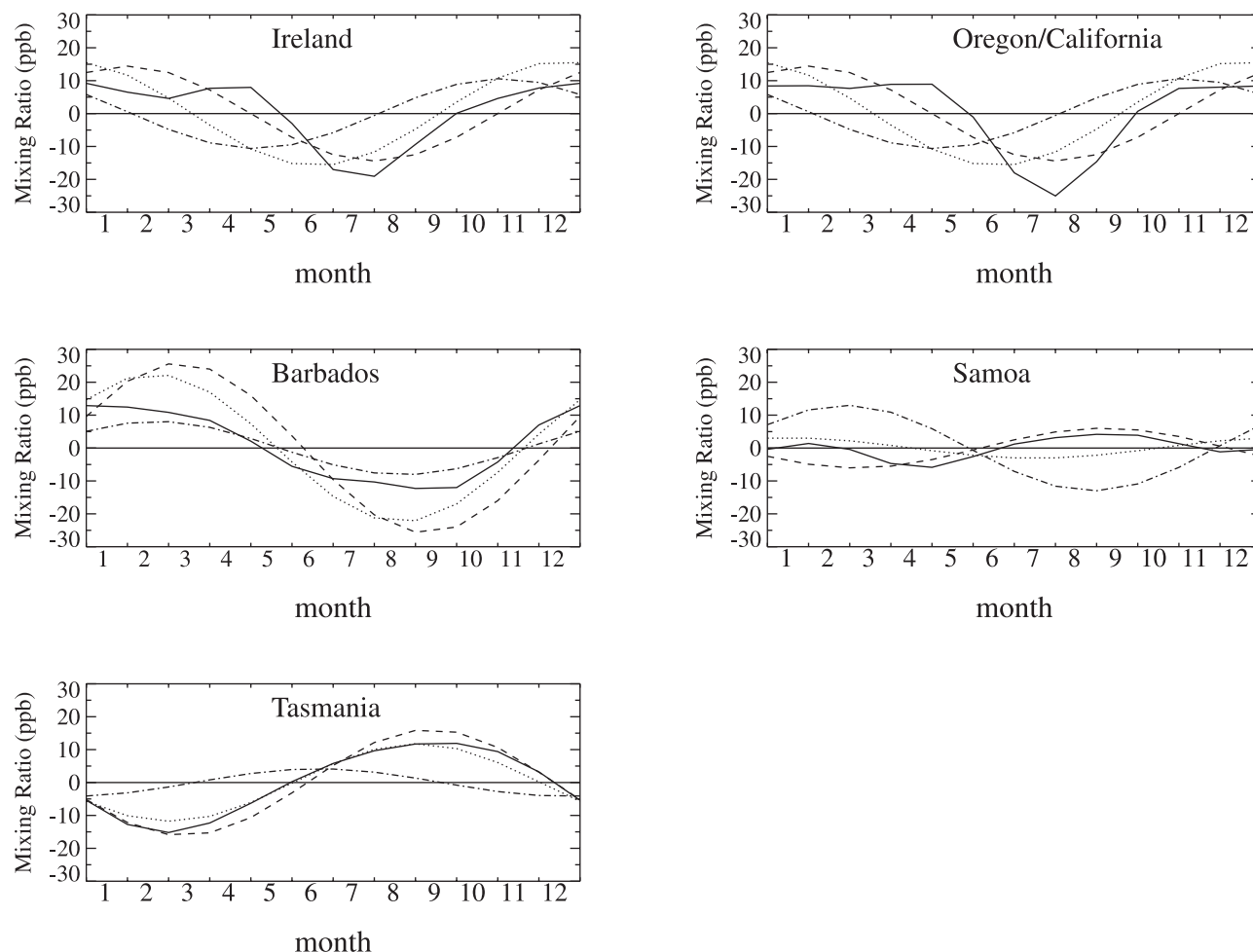


Figure 6. Observed and 2-D model calculated seasonal cycles in methane. The full line is the average of the GAGE/AGAGE and CMDL observed cycles from Table 1, the dash-dotted line is the model calculated 12-month cycle due to transport only, the dashed line is calculated using the combination of both transport and the OH cycle from *Spivakovsky et al.* [2000], and the dotted line also includes the effect of the seasonal cycle in emissions as given by the MOZART simulations.

the location of the Northern Hemisphere AGAGE sites, apart from the presence of month-to-month variability (note that sites west of Mace Head and the U.S. West Coast are shown (dash-dotted lines) in order to reduce regional pollution effects), are in reasonable agreement with the semihemispheric averages. At the Southern Hemisphere sites the simulated annual cycle at the sites, the semihemispheric surface averages and the 1000- to 500-mbar semihemispheric averages are all similar. Finally, Figure 7 also shows the calculated annual cycles (dotted lines) obtained with our 2-D model using MOZART OH values, the annually averaged transport rates which simulate the MOZART semihemispheric latitudinal gradient and with an annual cycle in transport which simulates the MOZART CCl_3F seasonal cycle. The calculation also includes the seasonal cycle in emissions used in MOZART. The 2-D model simulates the MOZART seasonal cycles in methane well except at Barbados (for which, as previously, indicated the 2-D transport is incorrect). Moreover, it clearly indicates

that MOZART's underestimation of the amplitude in the seasonal cycle at Tasmania relative to observation (see Figure 6) is caused by the underestimation of both the mean and the seasonal cycle in OH in 30°S – 90°S . Conversely, the successful simulation of the seasonal cycle at Tasmania shown in Figure 6 provides strong support for the *Spivakovsky et al.* [2000] (and similar *Prinn et al.* [2001]) OH values in 30°S – 90°S .

[50] Based on the strong dependence on OH and the minor role of transport and emissions in the seasonal cycle of methane at 30°S – 90°S , as demonstrated by Figures 6 and 7, the temporal variation of the seasonal cycle can be examined for information on changes in OH from 1986 to 2000. Figure 8a shows the time series of the calculated seasonal cycle amplitude for each year for both the GAGE/AGAGE data and the CMDL measurements at Cape Grim. Estimates for each year were made by fitting centered 2-year periods of monthly means with a linear trend and an annual cycle. The error bars are based on the

MOZART Results

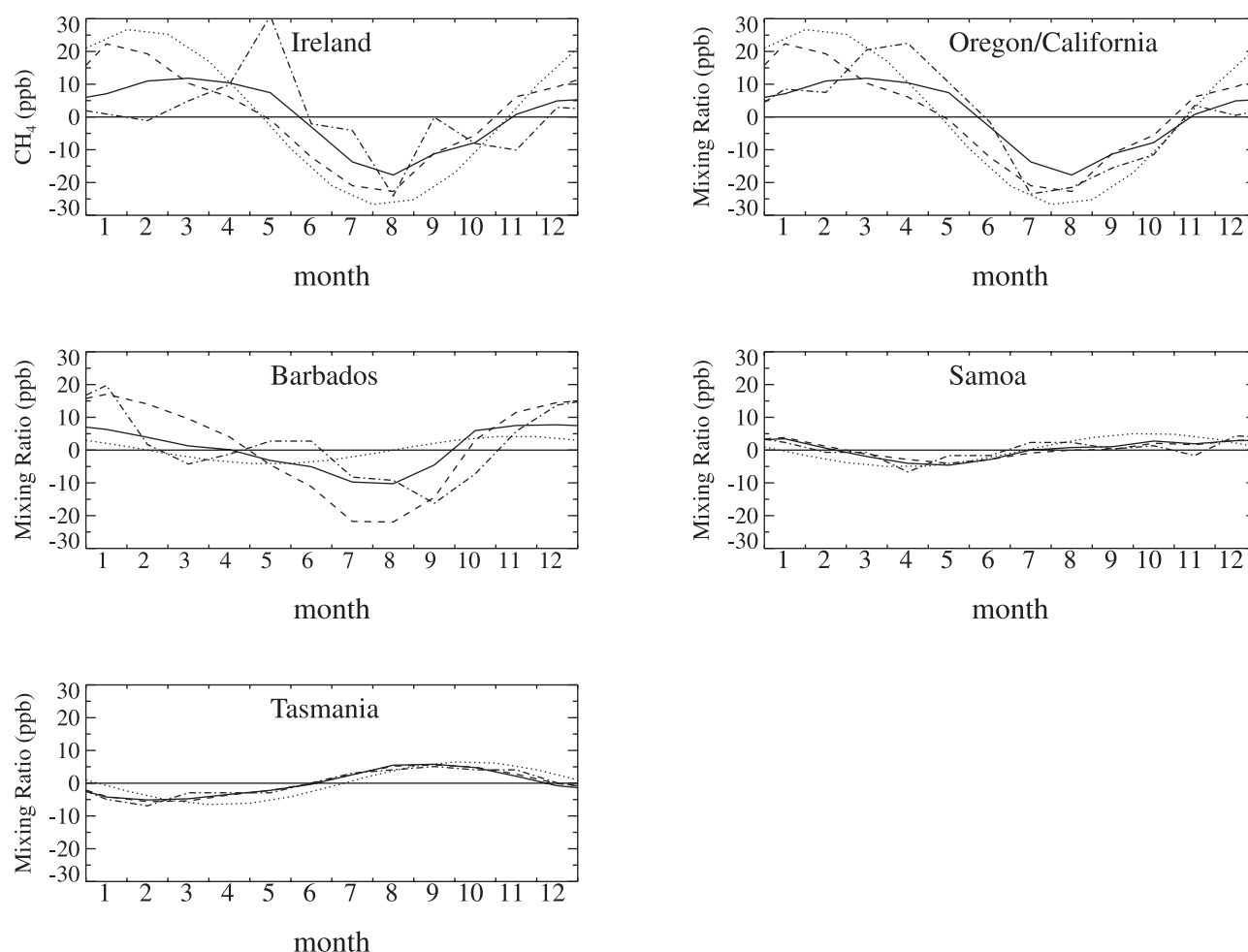


Figure 7. The semi-hemispheric seasonal cycles in methane from the MOZART simulations. The full line is for the semi-hemispheres from 1000–500 mbar, the dashed lines are semi-hemispheric surface averages, and the dash-dotted lines are for the AGAGE sites (with pollution adjustments). The dotted line is the 12-month periodicity calculated with the 2-D model using MOZART OH and seasonal cycles in emissions.

residuals from those fits. The error bars from GAGE observations are considerably larger than those from the AGAGE and CMDL measurements and in particular the anomalous value from the first 2 years of GAGE possess an especially large error bar. In estimating the trend in the annual cycle in Figure 8a (and Figure 8b) the GAGE/AGAGE and CMDL annual estimates of the seasonal cycle have been optimally combined (i.e., weighting each by the inverse of the residual variances). Figure 8b shows the same results as in Figure 8a except that the values are expressed as percentages of the annual means for each year. The mean annual cycle amplitude is 0.83% of the annual mean and the trend is $0.0003 \pm 0.0042\%/yr$ (95% confidence limits). It is the trend in this figure which is related to the trend in OH; an upward trend implies a decrease in OH. Dividing the trend by 0.83% and expressing the result in $\%/decade$ gives an OH trend for 1985–2000 for $30^\circ S$ – $90^\circ S$ of $-0.4 \pm 5.0\%/decade$ (95% confidence limits). This result is not significantly different

from the Southern Hemisphere OH trend shown by *Prinn et al.* [2001] which was derived from fitting the latitudinal gradient of CH_3CCl_3 as a function of time. The result is also similar to that from the shorter time series based estimate of *Dlugokencky et al.* [1997].

[51] In summary, it has been shown that transport variations make significant contributions to the seasonal cycle in methane except in $30^\circ S$ – $90^\circ S$. Thus the long-term trend in OH may be estimated from changes in the amplitude of the methane seasonal cycle in $30^\circ S$ – $90^\circ S$. The 2-D model has been shown to produce a good simulation of the observed 12-month cycles in methane in $0^\circ S$ – $90^\circ S$ using the *Spivakovsky et al.* [2000] or *Prinn et al.* [2001] OH distributions. On the other hand, it is difficult to simulate the observed cycle at Barbados with the 2-D model (and some 3-D models). In $30^\circ N$ – $90^\circ N$ it seems likely that the seasonal cycle in methane emissions is influencing the observed cycles at surface sites [see e.g., *Fung et al.*, 1991] but its determination from the measure-

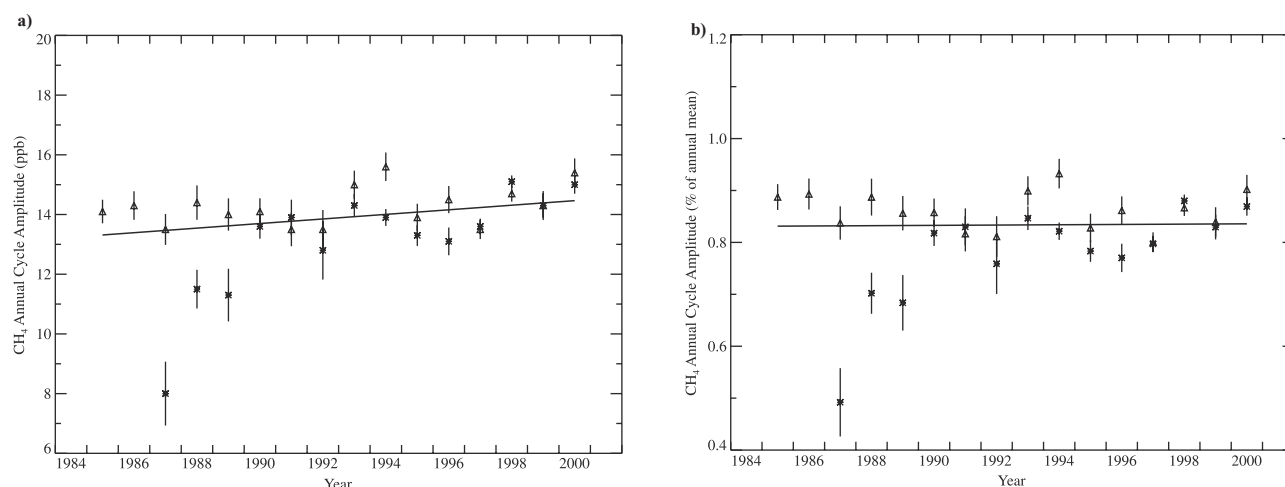


Figure 8. The time series of the amplitudes of the 12-month periodicities in methane measured at Cape Grim, Tasmania, from 1985 to 2000. The asterisks are from the GAGE/AGAGE data and the triangles are from the CMDL data; the error bars are the standard deviations of the residual after fitting centered 2-year periods with a 12-month cycle and a linear trend. The solid lines are the linear fit to the optimal combination of the GAGE/AGAGE and CMDL annual estimates. (a) Amplitudes expressed in ppb. (b) The 12-month cycle amplitudes expressed as percentages of the annual mean values.

ments is complicated by the need to accurately simulate the rate of mixing between the surface and the free troposphere.

7. Conclusions

[52] Methane measurements at the remote ground based GAGE/AGAGE sites since 1986 have been described. The short-term precision of the AGAGE measurements ranges from 2 ppb (0.1%) (since 1993) at Cape Grim, Tasmania, to 4 ppb (0.2%) at Mace Head, Ireland (since 1994). During GAGE the precisions were approximately a factor of 3 worse and the Mace Head measurements had somewhat larger baseline variability before 1992. Of particular value for assessing regional emissions is the short-term variability captured at several of the sites which is associated with regional pollution events [Ryall *et al.*, 2001; Dunse *et al.*, 2001].

[53] The GAGE/AGAGE measurements have been compared against the CMDL measurements at the same locations under baseline (i.e., free of local contamination) conditions. After removing for the calibration difference of approximately 1.2% between the two networks, the measurements occasionally show differences larger than 25 ppb even when the measurements were made within a few minutes of each other. More typically, however, there is excellent agreement between the CMDL and AGAGE measurements, especially the monthly means; the GAGE data are lower by approximately 5 ppb before 1989 and about 4 ppb larger from 1991 to 1995. This suggests that the short-term imprecision of the GAGE measurements applies equally to low-frequency variations.

[54] Using a 12-box 2-D model constrained by simultaneous measurements of methane, CCl_3F , CCl_2F_2 , and CH_3CCl_3 at just the five surface sites we infer net annual

global emissions of methane of approximately 545 Tg. Measurements from both networks at these sites indicate that the emissions were approximately constant (± 20 Tg) from 1985 to 1997 but that in 1998 the emissions increased by approximately 37 Tg. These estimates correspond to a lifetime for tropospheric methane due to its reaction with tropospheric hydroxyl of approximately 8.1 years. If, however, OH were to have changed over the past 2 decades as Prinn *et al.* [2001] have suggested, methane emissions would have been 20 Tg/yr higher before 1990, about 20 Tg/yr lower in 1997, and even lower thereafter.

[55] Using atmospheric transport rates determined from the CCl_3F and CCl_2F_2 observations at these same sites, the latitudinal distribution of methane emissions can be coarsely estimated using a 12-box 2-D model. The inferred latitudinal distribution of methane emissions is found to be significantly different from estimates by Fung *et al.* [1991], Hein *et al.* [1997] and others. Using the 3-D MOZART model, errors in the estimated spatial distribution of methane emissions have been traced to the use of 2-D transport rates derived from the CCl_3F and CCl_2F_2 measurements. The proportion of global emissions from each hemisphere is nevertheless robustly estimated, but the MOZART calculation indicates that the 2-D estimation procedure is shifting 5% of global CH_4 emissions from the Northern to the Southern Hemisphere. If the 2-D results are adjusted for the biased transport rates, the estimated Northern Hemisphere/Southern Hemisphere proportions of the emissions range from 0.73/0.27 using OH from Prinn *et al.* [2001] to 0.81/0.19 for the OH distribution used by both Hauglustaine *et al.* [1998] and Hein *et al.* [1997]. A corollary is that hemispheric emission ratios are expected to be accurately estimated using the 2-D model transport rates derived from the CFC measurements only for anthropogenic gases; for natural emissions possessing quite different spatial distribu-

tions, the estimates of the hemispheric emission percentages are mainly useful if errors of approximately 5% are insignificant relative to other sources of uncertainty.

[56] When the seasonal variations in the transport rates were tuned to produce agreement with the observed seasonal cycles in CCl_3F and CCl_2F_2 , the 12-month cycle in methane was found to be reasonably well simulated at all sites except Barbados. The problem is that it was not possible to reliably simulate the transport cycle at that site. It has been shown that the seasonal cycle of methane in 30°S – 90°S and at Cape Grim, Tasmania, is dominated by destruction by OH. We thus conclude that MOZART-1 [Hauglustaine et al., 1998] underestimates OH and the amplitude of its seasonal cycle by a factor of approximately 2 in this region, while the Spivakovsky et al. [2000] and Prinn et al. [2001] OH values in this region are approximately correct. From the trend in the GAGE/AGAGE and CMDL measurements of the seasonal cycle amplitude of methane we conclude that in 30°S – 90°S OH has been decreasing by $0\% \pm 5\%$ /decade from 1985 to 2000 (95% confidence limits).

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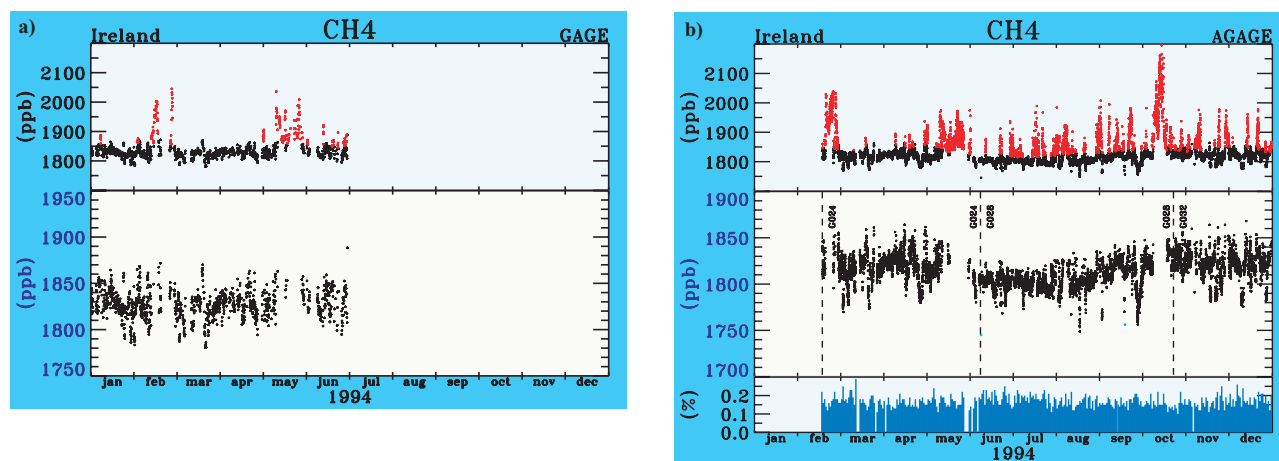


Figure 1. Methane measurements at Mace Head, Ireland, for 1994 (a) from the GAGE instrument, with a 2-hourly sampling interval and (b) from the GAGE instrument with a 40-min sampling interval. Red values indicate polluted values as designated by a statistically based pollution algorithm, standard tank changes are indicated by the dashed lines, and the blue values for AGAGE are the daily standard deviations of the measurements of the calibration air samples (i.e., measures of the instrumental precision).

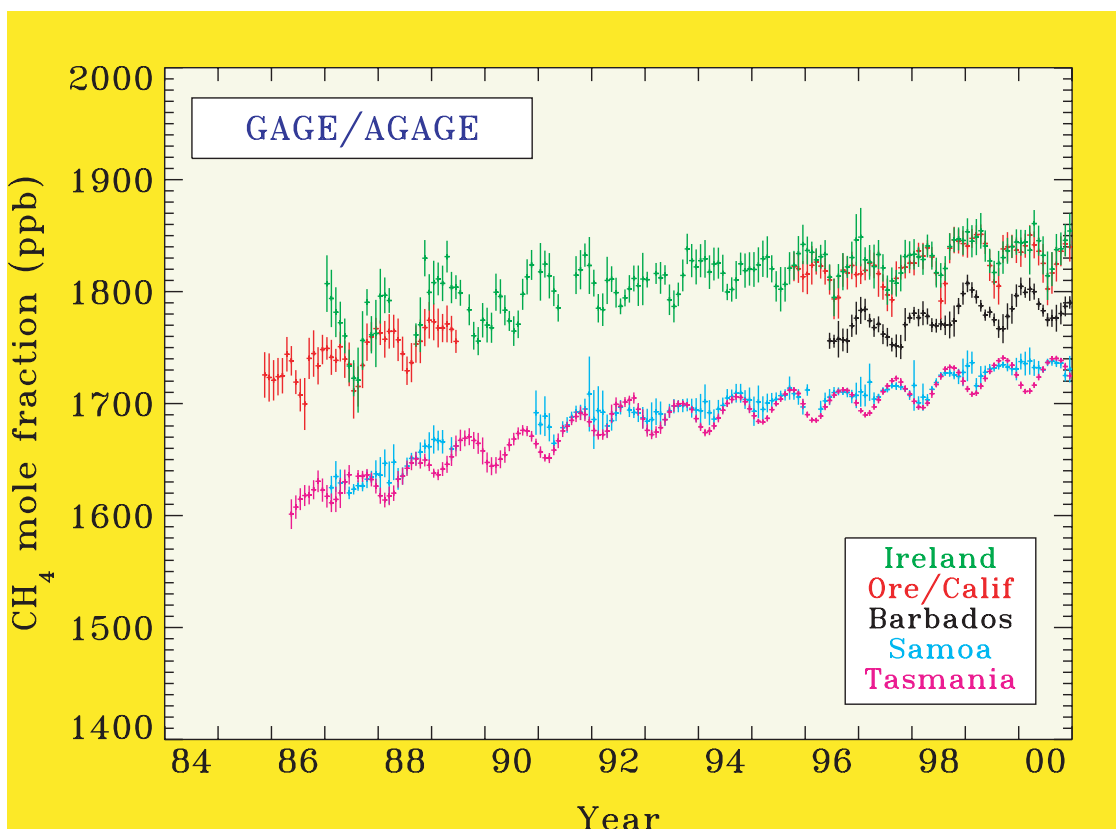


Figure 2. Monthly means and standard deviations of the baseline GAGE/AGAGE methane measurements at Mace Head, Ireland; Cape Meares, Oregon; and the follow-on site at Trinidad Head, California; Ragged Point, Barbados; Cape Matatula, Samoa; and Cape Grim, Tasmania.

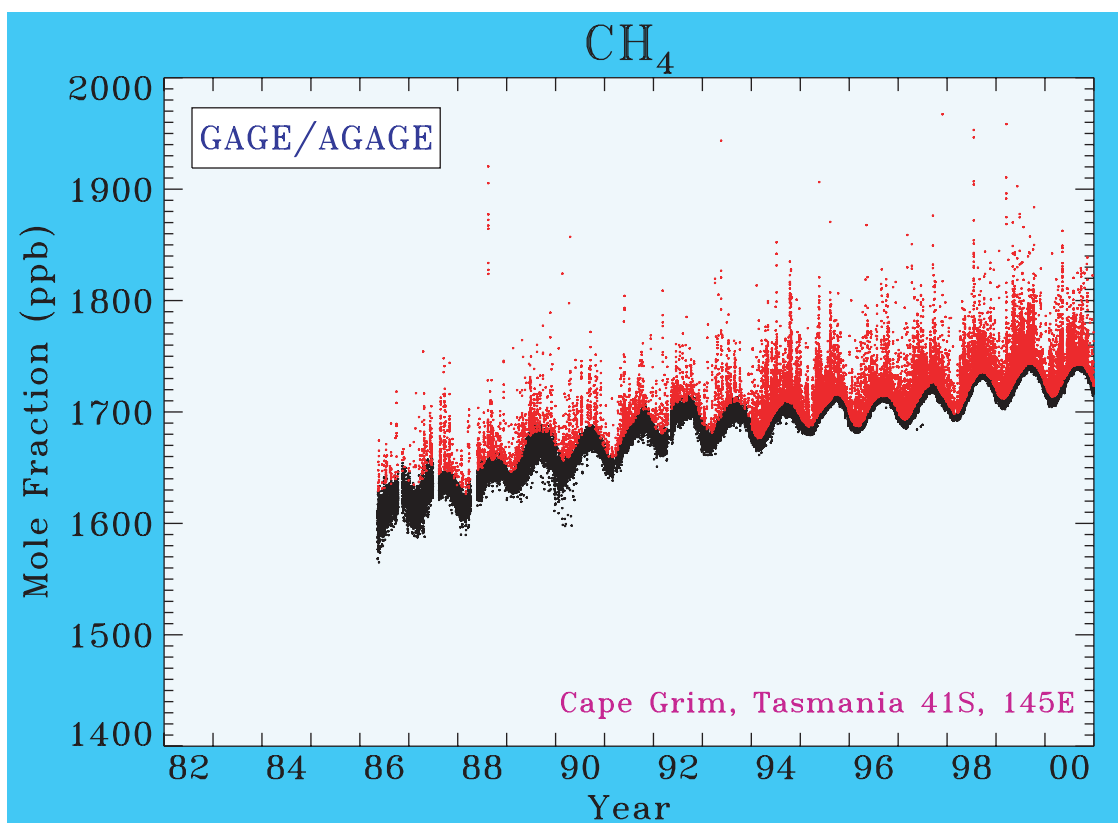


Figure 3. Every methane measurement by the GAGE and AGAGE instruments at Cape Grim, Tasmania. As in Figure 1, probably polluted values are indicated in red. The low values in early 1990 are probably related to measurement problems.